Lab 5 - Network Configuration and Troubleshooting

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This article is about configuring a network in the Ubuntu Client VM. It is focused on configuring the netplan file, testing the file, and refining any issues that might arise by using different troubleshooting commands. This article gives a step-by-step instruction guide, including all the commands and expected outputs of those commands.

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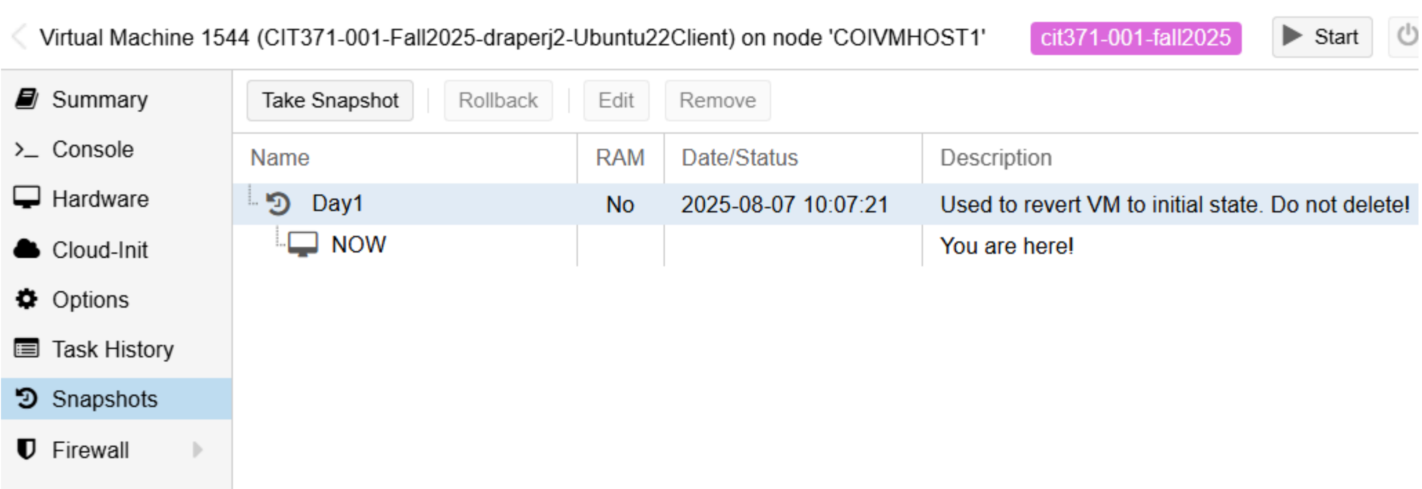
VM Snapshot(s)



Overview

The purpose of the Netplan utility is to provide a consistent, declarative way to configure networks on Ubuntu systems by using YAML files to define network settings. It is extremely beneficial to network administrators who need to configure and/or deploy complex networking setups in a consistent way across multiple servers. Ultimately, it is designed to make configurations easier and more descriptive, both for those doing the configuration, and for those following behind checking the configurations. Some of its key uses include defining IP addresses, network bridges, and applying configurations to physical and virtual network interfaces.

Part 3 Table

Command	Purpose
ip link show	This command checks if a network interface is up.
ip a	This command displays the assigned IP addresses.
ping -c 8.8.8.8	This command verifies connectivity, while also limiting the number of pings to 4.
nslookup microsoft.com	This command performs a DNS lookup and displays information about the Domain.
iftop	This command monitors real time network bandwidth usage.

Procedure

The first steps to take are to sign in to the Proxmox cluster, turn on the PFsense VM, then turn on the Ubuntu Client VM (Not the Server.)

Part 2: Network Configuration - Netplan

Once signed in to the Ubuntu Client we can begin the lab.

The first step is to open the Terminal and perform the following command:

sudo dhclient -r

This command releases the DHCP lease.

We then need to look up our Ubuntu client's network interface using this command:

ip -br a

The interface you are looking for will be attached to your 192.168.1.x (x representing your address) and will likely be *ens18* for example.

We then must switch to the root user so that we have elevated permissions, use this command:

sudo -i

For the password enter: *Student371!*

Next, we want to create a backup of the original netplan file before we make any changes; in order to do so we must change directories to the correct one and must find the name of the file. Use the following commands:

cd /etc/netplan

ls

cp 01-network-manager-all.yaml /root/01-network-manager-all.yaml.bak

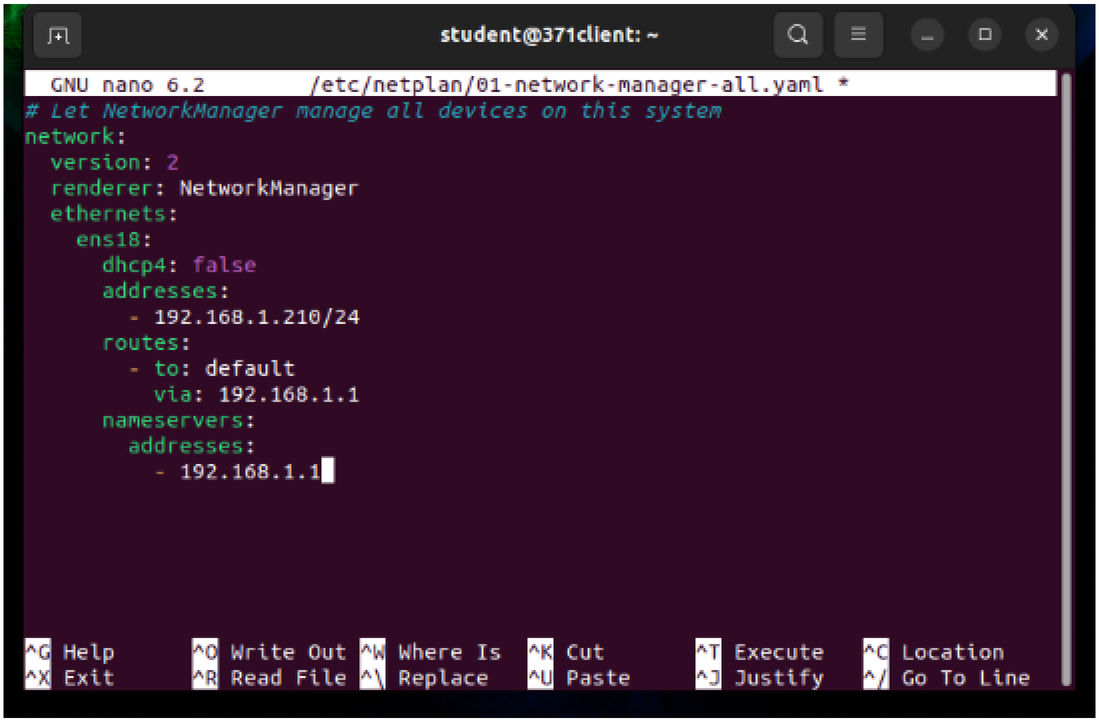
Next, we want to edit the file using nano, use the following command:

nano /etc/netplan/01-network-manager-all.yaml

Then enter the following information with the indentation (ensure that your interface (ens18) is correct):

```
network:
 version: 2
 renderer: NetworkManager
 ethernets:
   ens18:
     dhcp4: false
     addresses:
       - 192.168.1.210/24
     routes:
       - to: default
         via: 192.168.1.1
     nameservers:
       addresses:
         - 192.168.1.1
```

After writing, the file should look like this:



To save your configurations press Ctrl+O, press Enter, the press Ctrl+X.

Next, run the following command to apply the changes:

netplan apply

If you get a warning, just run this command:

sudo chmod 600 /etc/netplan/01-network-manager-all.yaml

netplan apply

Then reboot the system:

reboot

This completes Part 2 of the lab.

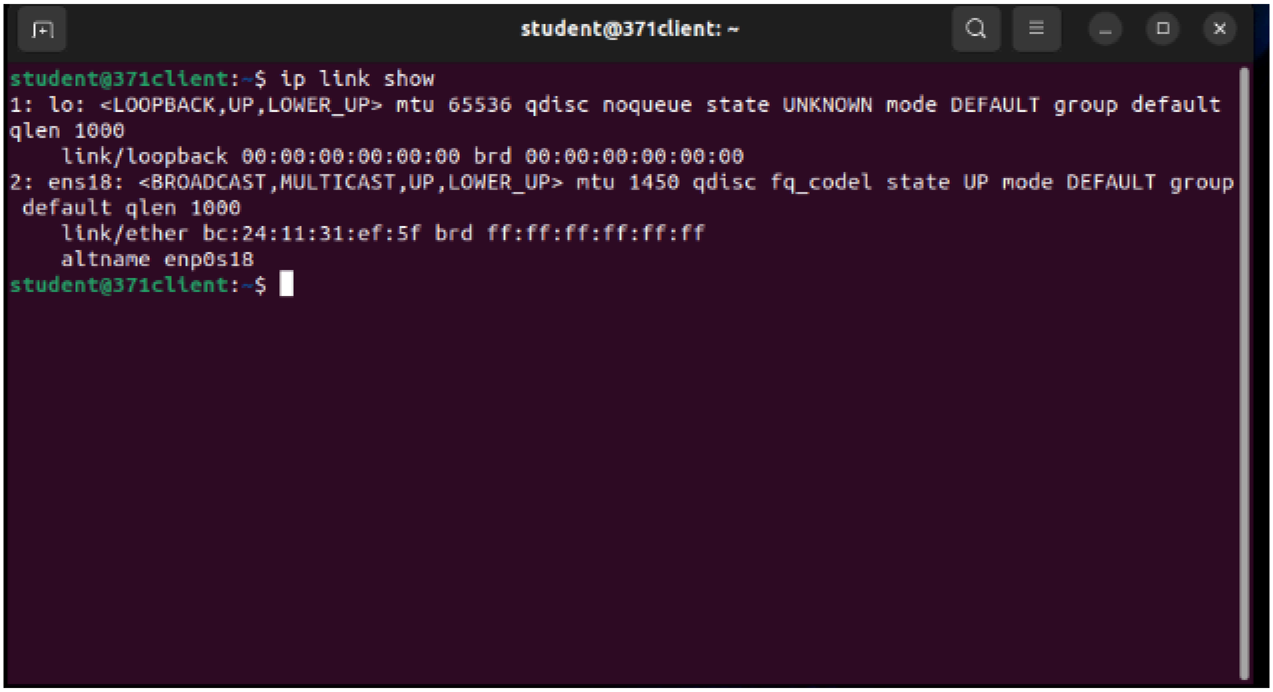
Part 3: Network Troubleshooting - Commands

Next, while in the Ubuntu Client, we will test our configurations.

First, we want to check to see if the network status is up/running. Use this command:

ip link show

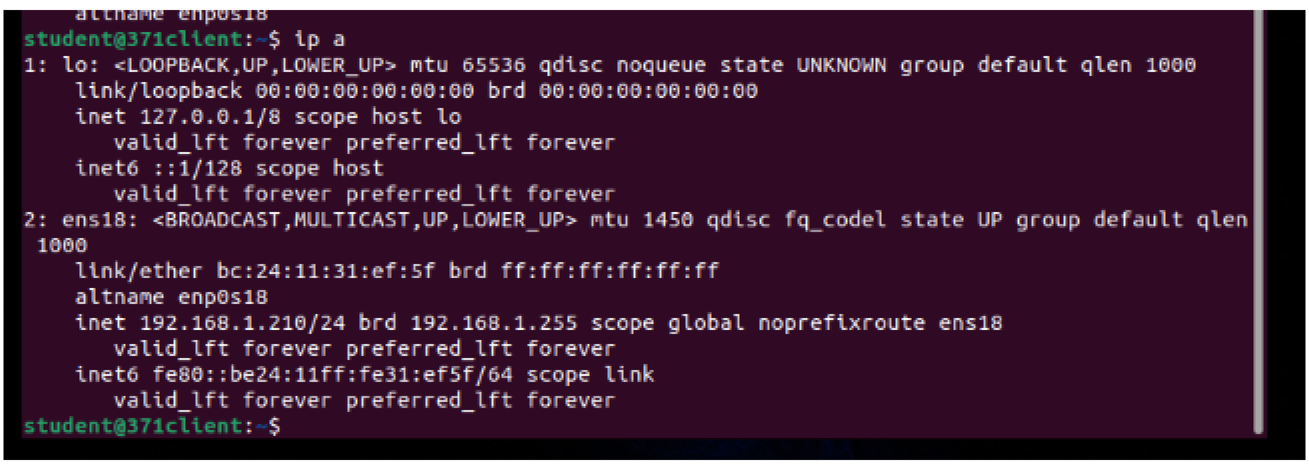
Your output should look like this:



Next, we want to verify that the client has the static IP address we assigned it earlier (192.168.1.210), use this command:

ip a

Your output should look like this:



Then we want to verify that the client has internet access by running the following command, ensuring we only pings the DNS server 4 times:

ping -c 4 8.8.8.8

Your output should look like this:

```
student@371client:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=7.67 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=7.53 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=7.68 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=7.63 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 7.532/7.628/7.680/0.059 ms
student@371client:~$
```

Next, we want to perform a DNS lookup on microsoft.com using the following command:

nslookup microsoft.com

Your output should look like this:

```
student@371client:~$ nslookup microsoft.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   microsoft.com
Address: 13.107.213.51
Name:   microsoft.com
Address: 13.107.246.51
Name:   microsoft.com
Address: 2603:1030:20e:3::23c
Name:   microsoft.com
Address: 2603:1030:c02:8::14
Name:   microsoft.com
Address: 2603:1020:201:10::10f
Name:   microsoft.com
Address: 2603:1010:3:3::5b
Name:   microsoft.com
Address: 2603:1030:b:3::152

student@371client:~$
```

After this, we want to install the *iftop* command, use the following commands:

sudo apt update

sudo apt install iftop -y

Then we want to run iftop using this command:

sudo iftop

Your output should look like this after running both commands:

```
student@371client:~$ sudo apt install iftop -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  iftop
0 upgraded, 1 newly installed, 0 to remove and 230 not upgraded.
Need to get 35.6 kB of archives.
After this operation, 95.2 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy/universe amd64 iftop amd64 1.0~pre4-7 [35.6 kB]
Fetched 35.6 kB in 0s (145 kB/s)
Selecting previously unselected package iftop.
(Reading database ... 204449 files and directories currently installed.)
Preparing to unpack .../iftop_1.0~pre4-7_amd64.deb ...
Unpacking iftop (1.0~pre4-7) ...
Setting up iftop (1.0~pre4-7) ...
Processing triggers for man-db (2.10.2-1) ...
student@371client:~$ sudo iftop
interface: ens18
IP address is: 192.168.1.210
MAC address is: bc:24:11:31:ef:5f
student@371client:~$
```

While this is running open firefox and search “Nintendo” with Google. Your output should look like this:

```
student@371client: ~
              1.91Mb          3.81Mb          5.72Mb          7.63Mb      9.54Mb
371client                   => is-content-cache-1.ps5.canonical    0b    284b     71b
                            <=                                     0b    324b     81b
371client                   => pfSense.home.arpa                   0b    136b   1.55Kb
                            <=                                   536b    330b   3.74Kb
371client                   => 82.221.107.34.bc.googleuserconte    0b     83b     83b
                            <=                                     0b     83b     83b
371client                   => ord38s32-in-f10.1e100.net           0b      0b   5.56Kb
                            <=                                     0b      0b   1.26Mb
371client                   => ord38s32-in-f4.1e100.net            0b      0b   17.9Kb
                            <=                                     0b      0b    402Kb
371client                   => ord38s31-in-f14.1e100.net           0b      0b   5.09Kb
                            <=                                     0b      0b    200Kb
371client                   => server-3-160-5-19.cmh68.r.cloudf    0b      0b   2.64Kb
                            <=                                     0b      0b    119Kb
371client                   => ord37s35-in-f3.1e100.net            0b      0b   3.46Kb
                            <=                                     0b      0b   23.6Kb
371client                   => ord37s33-in-f14.1e100.net           0b      0b   5.80Kb
                            <=                                     0b      0b   17.8Kb

TX:            cum:   674KB   peak:   506Kb             rates:     0b    503b   70.3Kb
RX:                  21.4MB           24.1Mb                     536b    737b   2.11Mb
TOTAL:               22.1MB           24.4Mb                     536b  1.21Kb   2.18Mb
```

Hit the ‘q’ key to exit this screen. Then use the exit command to exit. After you do this power off the Ubuntu Client and PFsense VM's. The lab has been finished successfully!

(All of the above screenshots were the ones requested for the labs.)

References

Ubuntu:

[About Netplan - Ubuntu Server documentation]

Akami TechDocs:

Network configuration using Netplan

LinuxConfig:

Netplan Ubuntu Configuration: A Beginner's Guide